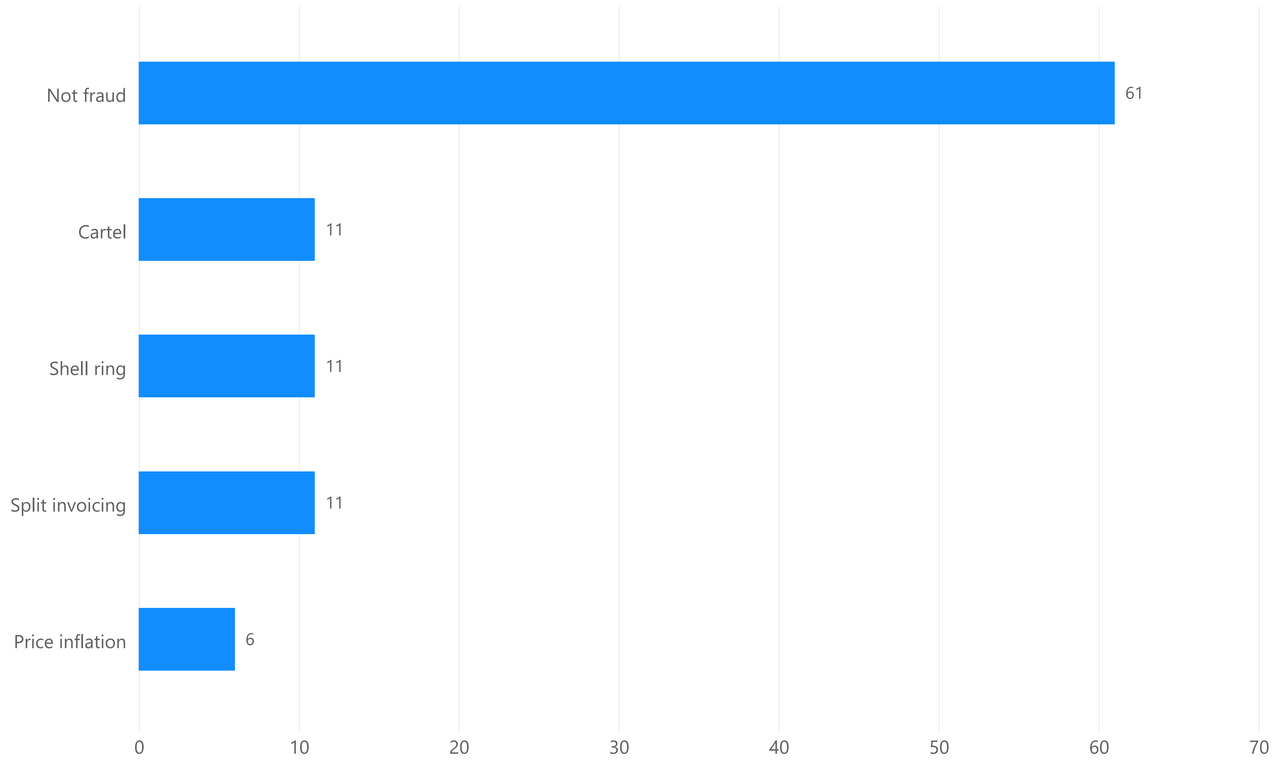
Top 10% of suppliers by risk score: 39 fraudulent suppliers found, 18 of them missed by audits.

Suppliers reviewed	Fraud caught	Confirmed by audit
100	39	21

Suppliers in queue

by True scheme (simulated)



Category

- ◯ Chemicals
- ◯ Electrical
- ◯ Electronic Components
- ◯ Facility Services
- ◯ Furniture
- ◯ Industrial Tools
- ◯ IT Hardware

Rist Rank	Supplier ID	Category	Risk score	Top risk drivers (SHAP)	Audit status	True Scheme
1	S03856	IT Hardware	0.79	p90 price ratio, median price ratio, top buyer share	Confirmed fraud	Price inflation
2	S02725	Electrical	0.77	near limit share, top buyer share, n buyers	Confirmed fraud	Split invoicing
3	S04780	Chemicals	0.74	top buyer share, shared address count, n buyers	Confirmed fraud	Shell ring
4	S04747	IT Hardware	0.73	n bids, n buyers, close loss share	Not confirmed	Cartel
5	S00143	Furniture	0.72	n bids, n buyers, top buyer share	Not confirmed	Cartel
6	S00127	IT Hardware	0.72	near limit share, n buyers, burst share	Confirmed fraud	Split invoicing
7	S00630	Packaging	0.71	near limit share, shared director count, n buyers	Not confirmed	Not fraud
8	S00301	Furniture	0.71	p90 price ratio, n bids, n buyers	Not confirmed	Cartel
9	S04369	IT Hardware	0.71	near limit share, n buyers, top buyer share	Confirmed fraud	Split invoicing
10	S02919	Electrical	0.70	n bids, n buyers, close loss share	Not confirmed	Cartel
11	S03115	Logistics	0.69	p90 price ratio, median price ratio, n buyers	Confirmed fraud	Price inflation
12	S04511	IT Hardware	0.69	n bids, n buyers, close loss share	Not confirmed	Cartel
13	S01315	Chemicals	0.68	top buyer share, shared address count, n buyers	Not confirmed	Shell ring
14	S03186	Industrial Tools	0.68	near limit share, shared director count, n buyers	Confirmed fraud	Split invoicing
15	S04714	Furniture	0.67	near limit share, p90 price ratio, top buyer share	Confirmed fraud	Split invoicing
16	S01026	Furniture	0.64	n bids, n buyers, top buyer share	Not confirmed	Cartel
17	S00511	Logistics	0.64	near limit share, n buyers, burst share	Confirmed fraud	Split invoicing
18	S00292	Electrical	0.63	top buyer share, n buyers, unshipped share	Not confirmed	Shell ring
19	S00606	Furniture	0.63	near limit share, n buyers, burst share	Not confirmed	Not fraud
20	S03908	Packaging	0.62	p90 price ratio, median price ratio, n buyers	Not confirmed	Price inflation
21	S02530	Packaging	0.62	shared director count, top buyer share, n buyers	Not confirmed	Not fraud
22	S00988	Safety Equipment	0.62	n bids, n buyers, close loss share	Confirmed fraud	Cartel
23	S03175	Furniture	0.62	n bids, n buyers, close loss share	Not confirmed	Cartel
24	S03026	Office Supplies	0.61	shared director count, top buyer share, n buyers	Confirmed fraud	Shell ring
25	S03919	IT Hardware	0.61	p90 price ratio, n bids, top buyer share	Not confirmed	Cartel
26	S02488	Steel & Metals	0.61	p90 price ratio, n bids, n buyers	Not confirmed	Not fraud
27	S01088	Electrical	0.61	shared director count, top buyer share, n buyers	Not confirmed	Not fraud
28	S03666	Logistics	0.60	top buyer share, n buyers, unshipped share	Not confirmed	Not fraud
29	S04478	IT Hardware	0.60	shared director count, top buyer share, n buyers	Not confirmed	Not fraud
30	S03179	Logistics	0.60	near limit share, n buyers, burst share	Not confirmed	Not fraud
31	S02549	Steel & Metals	0.60	p90 price ratio, near limit share, median price ratio	Not confirmed	Not fraud
32	S01227	Packaging	0.60	p90 price ratio, top buyer share, n buyers	Not confirmed	Not fraud
33	S00931	Chemicals	0.59	near limit share, n buyers, burst share	Confirmed fraud	Split invoicing
34	S03212	Safety Equipment	0.59	n bids, n buyers, close loss share	Confirmed fraud	Cartel
35	S01951	Facility Services	0.59	shared address count, top buyer share, n buyers	Not confirmed	Shell ring
36	S04064	Facility Services	0.58	n bids, n buyers, close loss share	Not confirmed	Cartel

Metric

- ○ PR-AUC
- ○ Precision at top 10%
- ◉ Recall at top 10%
- ○ Recall on fraud audits missed
- ○ ROC-AUC

Recall per fraud scheme at a top-10% review budget

model	Cartels	Price inflation	Shell rings	Split invoicing
GraphSAGE	0.74	0.55	0.69	0.61
Hybrid (XGBoost + GraphSAGE)	0.75	0.62	0.73	0.77
Neural net (no graph)	0.77	0.60	0.63	0.58
XGBoost (+ relational)	0.81	0.61	0.65	0.89
XGBoost (tabular)	0.64	0.51	0.51	0.87

Average of mean by model

Score by model (metric selected on the left)



Mean over 5 train/test splits, graded on simulated ground truth. Review budget: top 10% of suppliers.

Shell rings and cartels look normal on their own invoices, relational features (shared bank accounts, directors and bidding partners) are what expose them.

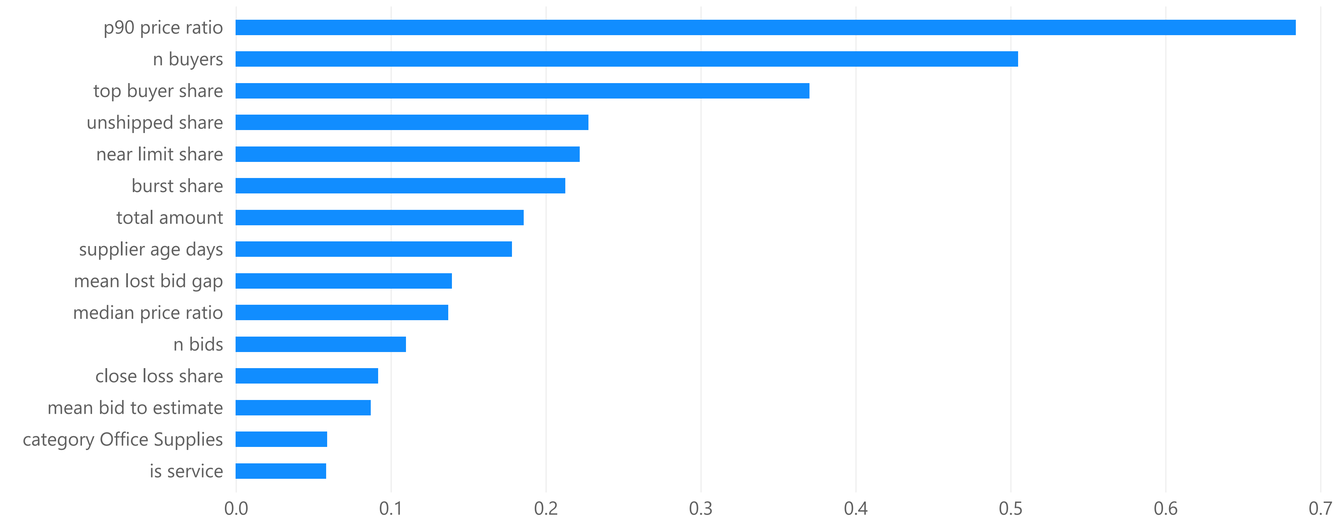
model

Hybrid (XGBoost + GraphSAGE)

XGBoost (tabular)

XGBoost (+ relational)

What drives the risk score (mean absolute SHAP value)



Price and invoicing patterns by supplier (true scheme simulated)

